2594. Minimum Time to Repair Cars

Solved

Medium Topics Hint

You are given an integer array `ranks` representing the **ranks** of some mechanics. $ranks_i$ is the rank of the ith mechanic. A mechanic with a rank `r` can repair n cars in $r * n^2$ minutes.

You are also given an integer `cars` representing the total number of cars waiting in the garage to be repaired.

Return *the **minimum** time taken to repair all the cars.*

Note: All the mechanics can repair the cars simultaneously.

Example 1:

> **Input:** ranks = [4,2,3,1], cars = 10
> **Output:** 16
> **Explanation:**
> - The first mechanic will repair two cars. The time required is 4 * 2 * 2 = 16 minutes.
> - The second mechanic will repair two cars. The time required is 2 * 2 * 2 = 8 minutes.
> - The third mechanic will repair two cars. The time required is 3 * 2 * 2 = 12 minutes.
> - The fourth mechanic will repair four cars. The time required is 1 * 4 * 4 = 16 minutes.
> It can be proved that the cars cannot be repaired in less than 16 minutes.

Example 2:

> **Input:** ranks = [5,1,8], cars = 6
> **Output:** 16
> **Explanation:**
> - The first mechanic will repair one car. The time required is 5 * 1 * 1 = 5 minutes.
> - The second mechanic will repair four cars. The time required is 1 * 4 * 4 = 16 minutes.
> - The third mechanic will repair one car. The time required is 8 * 1 * 1 = 8 minutes.
> It can be proved that the cars cannot be repaired in less than 16 minutes.

Constraints:

- `1 <= ranks.length <= 10^5`
- `1 <= ranks[i] <= 100`
- `1 <= cars <= 10^6`

Seen this question in a real interview before? 1/6

Yes No

Accepted 161,012/271.9K | Acceptance Rate 59.2%

Topics

Hint 1

Hint 2

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